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Department of Computer Science & IT

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Q Define Buffer solution.

Ans: A buffer solution is an aqueous solution that can resist significant changes in pH when small amounts of an acid or a base are added to it. It is typically composed of a weak acid and its conjugate base or a weak base and its conjugate acid.

Q Write the principle of fluorescence spectroscopy.

Ans: The principle is based on fluorescence, a process where a molecule, known as a fluorophore, absorbs light at a specific wavelength and is promoted to a higher energy state. It then quickly returns to its ground state by emitting light at a longer, lower-energy wavelength. The intensity and wavelength of this emitted light provide information about the environment.

Q Write down the component of fluorescence microscope

Ans: The essential components of a fluorescence microscope are:

- Light Source
- Excitation filter
- Dichroic Mirror
- Objective Lens
- Emission Filter
- Detector

Q Name the different stages of sterilization methods.

Ans: The main stages are:

- Heat Sterilization
- Chemical Sterilization
- Radiation Sterilization
- Sterile Filteration

Q How buffer solution and pH can correlate

Ans: Buffer solutions and pH are directly correlated because a buffer's entire function is to maintain a stable pH. The Henderson-Hasselbalch equation mathematically describe this relationship:

$$pH = pK_a + \log\left(\frac{[A^-]}{[HA]}\right)$$

Q Explain the Role of Buffer Solution in human body giving proper mechanism.

Ans Buffers are critical in the human body to maintain homeostasis. The most important one is the bicarbonate buffer system, which keep the pH of human blood stable at approximately 7.4.

- Components: The buffer consists of carbonic acid (H_2CO_3), the weak acid and the bicarbonate ion (HCO_3^-, the conjugate base).
- Mechanism: If the blood becomes too acidic, the bicarbonate ions act as a base and neutralize the acid:

$$H^+ + HCO_3^- \rightarrow H_2CO_3$$

(The resulting carbonic acid is then converted into CO_2 and exhaled by the lungs)

- If the blood becomes too alkaline (too few H^+ ions), the carbonic acid acts as an acid, donating protons to restore the pH:

$$H_2CO_3 \rightarrow H^+ + HCO_3^-$$

This dynamic equilibrium ensures that blood pH remains within a very narrow, life sustaining range.

Q Briefly explain different phases of the bacterial growth curve.

Ans: The bacterial growth curve illustrates the predictable pattern of bacterial population growth over time in a closed system. It has four distinct phases:

(i) Lag Phase: After inoculation into new media, bacteria are metabolically active but not dividing. They are adapting to the new environment, synthesizing enzymes, and preparing for growth.

(ii) Log (Exponential Phase): Bacteria undergo rapid, exponential cell division. The population doubles at a constant rate. This is the phase where bacteria are healthiest and most active.

(iii) Stationary Phase: The rate of cell division equals the rate of cell death. Growth slows down because nutrients are becoming depleted, and toxic waste products are accumulating. The total number of viable bacteria remains constant.

(iv) Death (Decline) Phase: The rate of cell death exceeds the rate of cell division. Conditions become increasingly hostile, leading to a sharp decline in the number of living bacteria.

Q Explain about the measuring mechanical strength of cell.

Ans Measuring the mechanical strength of a single cell helps us understand its health and function. Key techniques include:

- Atomic Force Microscopy (AFM) - This is a very high-resolution technique where a sharp tip on a flexible cantilever is used to "poke" or indent the cell surface. By measuring how much the cantilever bends for a given force, scientists can calculate the cell's stiffness (Young's Modulus).
- Micropipette Aspiration: A tiny glass pipette applies a small amount of suction to the cell surface. The amount of the cell deforms and is pulled into the pipette for a given pressure is measured to determine its mechanical properties.
- Optical Tweezers: Uses highly focused laser beams to create an "optical trap" that can hold and stretch a cell. The force required to stretch the cell by a certain amount reveals its elasticity and deformability.

Q Why buffer solutions are important in biological systems.

Ans: Buffer solutions are fundamentally important in biological systems to maintain a constant internal pH, a state known as Homeostasis.

* Optimal Enzyme Function - Nearly all biochemical reactions are catalyzed by enzymes, which are highly sensitive to pH. Most enzymes have a very narrow optimal pH range. If the pH changes, the enzyme can lose its shape and its activity, disrupting vital metabolic pathways.

* Cellular and Bodily Integrity: Buffers in blood, cells, and tissues prevent drastic pH shifts that could damage cellular structure and lead to death.

Q Distinguish between biosafety level 1, 2, 3 & 4.

Ans

Feature	BSL-1	BSL-2	BSL-3	BSL-4
Agents handled	Non-pathogenic organisms	Moderate risk pathogens causing mild disease	Serious pathogens that may cause lethal disease, airborne transmission possible	Highly dangerous pathogens, often fatal, no treatment or vaccine
Risk level	Minimal	Moderate	High	Extreme
Containment	Basic laboratory practices	Limited access, biohazard warning signs	Controlled access, directional airflow, sealed lab	Maximum containment isolated facility, airtight lab.
PPE (Personal Protective Equipment)	Lab coat, gloves, eye protection as needed	Lab coat, gloves face shield / goggles	Protective clothing, gloves, respirator	Full-body positive-pressure suit with dedicated air supply.
Facility design	Open bench work, sink for handwashing	Biosafety cabinets for infectious aerosols, autoclave available	Biosafety cabinets specialized ventilation	Isolated building/zone, Class III biosafety cabinet or positive-pressure suits, dedicated exhaust & decontamination systems.
Examples of work	Teaching labs, routine microbiology	Clinical diagnostics, human/animal pathogenes of moderate risk	Research on airborne pathogens, vaccine development	Research on deadly viruses with no cure maximum security labs.

Q How fluorescence microscopy works? Write the principle of fluorescence spectroscopy, Explain electron transition in fluorescence spectroscopy.

Ans How Fluorescence Microscopy Works: A high-intensity light source is used, but a specific excitation filter only allows the desired wavelength of light to pass. The light is reflected by a dichroic mirror onto the specimen, exciting fluorescence molecules within it. These molecules then emit light at a longer wavelength. This emitted light passes through the dichroic mirror and a second filter, which blocks any residual excitation light. The purified emitted light is then captured by a detector to create a high-contrast image of the fluorescently labeled structures.

* Principle of Fluorescence Spectroscopy:

- Fluorescence spectroscopy is based on the principle that certain molecules absorb light at a specific wavelength and then emit light at a longer wavelength.
- The difference in wavelength between excitation and emission is called the Stokes shift.
- By analyzing the emission spectra, one can identify, quantify, and study the environment of fluorophores.
- The technique is widely used in biology, chemistry, and material sciences to detect trace amounts of analytes and study molecular interactions.

* Electron Transition In Fluorescence Spectroscopy:

(i) Ground State (S_0):

- The electron starts in the lowest energy state

(ii) Excitation ($S_0 \rightarrow S_1$ or S_2):

- Absorption of a photon excites the electron to a higher electronic state.

(iii) Vibrational Relaxation:

- The excited electron loses some energy through non-radiative processes, settling into the lowest vibrational level of the excited state.

(iv) Fluorescence Emission ($S_1 \rightarrow S_0$):

- The electron ~~must~~ returns to the ground state, releasing energy as a photon
- This emitted photon has lower energy and lower wavelength than the absorbed photon.

(v) Other Pathways:

- Competing process include internal conversion, intersystem crossing or non-radiative decay.

Q Discuss about the sterilization various techniques used in Laboratory.

Ans: Sterilization is the process of completely eliminating or destroying all forms of microbial life, including spores. The main techniques used in a laboratory are:

- Moist Heat Sterilization (Autoclave): This is the most common and reliable method. It uses steam under high pressure to denature proteins and

kill all microorganisms. It's used for sterilizing culture media, glassware, and surgical instruments.

* Dry Heat Sterilization (Hot Air Oven) - This method uses high temperatures for a longer duration. It kills microbes by oxidation. It is used for materials that are sensitive to moisture but can withstand high temperatures, such as glassware, metal instruments, and powders.

* Filtration - This technique physically removes microbes from heat-sensitive liquids. The liquid is passed through a filter with pores that are too small for bacteria to pass through. It does not remove viruses effectively.

* Radiation Sterilization:

- UV Radiation → Uses non-ionizing radiation to damage microbial DNA. It has poor penetration and is primarily used for surface sterilization, such as inside biosafety cabinets and for decontaminating air and water.

- Ionizing Radiation (Gamma Rays) → Uses high-energy radiation to destroy DNA. It is highly penetrating and is used for sterilizing pre-packaged disposables like syringes, petri dishes, and gloves.

* Chemical Sterilization: Uses toxic chemicals to kill microbes.

- Gases (e.g., Ethylene Oxide) - A highly effective gas used for sterilizing heat-sensitive items like plastic tubing and electronic devices. It is toxic and requires a long aeration period after sterilization.

- Liquids (e.g., Glutaraldehyde Bleach) :-
Used for disinfecting surfaces and soaking heat-sensitive instruments.